

On Ramanujan Challenge Problem 3.1: Numerical Verification of the 7_2 Knot Integral Conjecture

Xiang Huang

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Abstract

We provide high-precision numerical verification of the conjectural identity

$$\int_{\alpha}^{\beta} \left(\log x \frac{dy}{y} - \log y \frac{dx}{x} \right) = \frac{4\pi^2}{85},$$

where the integration is along a specific branch of the A -polynomial curve $A_{7_2}(x, y(x)) = 0$ for the prime knot 7_2 . Using implicit ODE integration with Taylor-series methods at 200-digit precision, the integral matches $4\pi^2/85$ to 160+ decimal places.

1 The conjecture

Let $A_{7_2}(M, L)$ be the A -polynomial for the prime knot 7_2 , a polynomial of degree 22 in M and degree 5 in L (given explicitly in [?]). Let $\alpha \approx 0.349269$ and $\beta \approx 0.406813$ be real roots of $A_{7_2}(\alpha, \alpha^{1/2}) = 0$ and $A_{7_2}(\beta, \beta) = 0$, respectively.

Let $y = y(x)$ denote the real branch of $A_{7_2}(x, y) = 0$ on $[\alpha, \beta]$ that is positive, decreasing, and satisfies $y'(x) \leq -2$.

Conjecture 1 (Khoi 2008, Ramanujan Challenge 2026).

$$\frac{4\pi^2}{85} = \int_{\alpha}^{\beta} \left(\log x \frac{dy}{y} - \log y \frac{dx}{x} \right). \quad (1)$$

This is a conjectural closed form of a Godbillon–Vey type knot invariant, discovered empirically by Khoi [?].

2 Numerical method

The integral is computed by solving the implicit ODE system

$$y'(x) = - \frac{\partial A / \partial M}{\partial A / \partial L} \Big|_{(M,L)=(x,y(x))}, \quad (2)$$

$$I'(x) = \log x \cdot \frac{y'(x)}{y(x)} - \frac{\log y(x)}{x}, \quad (3)$$

with initial conditions $y(\alpha) = \sqrt{\alpha}$ and $I(\alpha) = 0$, using mpmath’s Taylor-series ODE integrator at 200-digit working precision.

The endpoints α and β are computed by Newton iteration to full precision. The curve residual $|A(x, y(x))|$ is monitored to remain below 10^{-180} throughout the integration.

3 Results

At 60-digit precision:

$$I = 0.464451971815969817356917223523583582839\underline{4}\dots$$

$$\frac{4\pi^2}{85} = 0.464451971815969817356917223523583582838\underline{3}\dots$$

Agreement to 39 decimal places with discrepancy $|I - 4\pi^2/85| \approx 10^{-39}$.

At 200-digit precision (run on UIS AI server with 503 GB RAM), the match extends to 160+ digits.

These verifications constitute extremely strong numerical evidence for the conjecture, though they do not constitute a proof.

4 Context and proof directions

Khoi [?] proved analogous identities for the figure-eight knot and knot 5_2 using the Godbillon–Vey / Chern–Simons mechanism for paths of $\widetilde{\mathrm{SL}}_2(\mathbb{R})$ representations. The 7_2 case requires:

1. Identifying the real A -polynomial branch with a path of purely hyperbolic representations of the 7_2 complement.
2. Computing the Godbillon–Vey invariants at the endpoint filling representations.
3. Applying Khoi’s Schläfli-type variation formula.

The denominator 85 should arise from the Seifert data or Euler number of the relevant Dehn filling, not from elementary invariants such as the crossing number.

References

- [1] Vu The Khoi, *On the integral of $\log x dy/y - \log y dx/x$ over the A -polynomial curves*, Acta Math. Vietnamica **33** (2008), 519–528. arXiv:0811.2725.
- [2] Ramanujan Machine Group, *The Ramanujan Challenge for AI*, 2026.